

Garvit Sonawala

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RESEARCH INTERESTS

Theoretical and computational particle physics and cosmology, with interests spanning dark-sector model building, dark matter and early-Universe cosmology, gravitational-wave phenomenology, and flavour physics. Particularly interested in connecting particle-physics models to cosmological and observational signatures.

EDUCATION

BS-MS (Integrated) in Physics, IISER Thiruvananthapuram 2022 – 2027 (expected)
CGPA: 7.8 / 10

BS in Data Science and Applications, IIT Madras 2023 – present
CGPA: 8.25 / 10

CBSE Class XII, Delhi Public School, Varanasi (92.4%) 2021

RESEARCH EXPERIENCE

Master's Thesis: Rare B Decays as a Probe of New Physics, IIT Gandhinagar May 2026 – present
Supervisor: Dr. Rusa Mandal

Studying loop-level interactions of BSM extensions and matching the underlying UV theory onto the weak effective theory (WET) to extract Wilson coefficients, confronting the resulting predictions against current LHCb and Belle data on $b \rightarrow s \ell^+ \ell^-$ transitions to narrow the observed deviation from the Standard Model. Presently focused on the exclusive decay $B \rightarrow K^* \ell^+ \ell^-$.

Research Intern, Tata Institute of Fundamental Research, Mumbai May – July 2025
Supervisor: Dr. Achamveedu Gopakumar

Modelled pulsar-timing-array residuals from circular supermassive black-hole binaries to 2PN accuracy using 12-year PTA datasets, and trained convolutional, recurrent and Bayesian neural networks to classify waveforms and recover chirp mass in the low signal-to-noise regime.

Research Intern, IIT Guwahati July – September 2024
Supervisor: Dr. Subhaditya Bhattacharya

Derived the Boltzmann equation governing the dark matter yield in a radiation-dominated Universe, analytically characterising thermal freeze-out and the scaling of the relic abundance with the annihilation cross section.

Minor Research Project, Computational Chemistry, IISER Thiruvananthapuram Jan – May 2026
Supervisor: Dr. R. S. Swathi

Implemented a particle-swarm optimisation code with ring topology and L-BFGS-B local refinement for global structure prediction of $(B_{12})_n$ clusters ($n = 2-10$), using improved Lennard-Jones and antiaromatic interaction models. Benchmarked against published global minima via HPC parameter sweeps.

PRESENTATIONS

Rare Decay $B \rightarrow K^* \ell^+ \ell^-$: From Effective Field Theory to New Physics SRIP 2026, IIT Gandhinagar
Poster: one-loop calculation, EFT matching, hadronic uncertainties, angular observables, and the tension in P_5' .

Exploring Dark Matter Freeze-Out ANVESH 2024, IISER Thiruvananthapuram
Poster: thermal freeze-out, relic abundance, and the observational case for dark matter.

HONORS & AWARDS

Qualified GATE (Physics, PH), 2026.

TECHNICAL SKILLS

Scientific computing: Mathematica (FeynArts, FeynCalc, Package-X), Python (NumPy, SciPy, Matplotlib, Jupyter), L^AT_EX, Git, Linux
Machine learning: PyTorch, scikit-learn; convolutional, recurrent and Bayesian neural networks
Also familiar with: Java, SQL, Flask (IIT Madras data science coursework)

RELEVANT COURSEWORK

Physics: Quantum Field Theory, Particle Physics, Quantum Mechanics I & II, Statistical Mechanics, Classical Mechanics, Electrodynamics, Special & General Relativity, Astrophysics, Nuclear Physics

Mathematics & Computation: Linear Algebra, Multivariable Calculus, Probability & Statistics, Machine Learning

Additional: Electroweak Interactions in the Standard Model, NPTEL (Dr. Subhaditya Bhattacharya, IIT Guwahati)

REFERENCES

Dr. Rusa Mandal, IIT Gandhinagar
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LEADERSHIP

Core Committee Member, PSIT Physics Society, IISER TVM Aug 2024 – Aug 2025